

The Estimator Decides:

What Curricula and Failure Recycling Can and Cannot Do in RL with Verifiable Rewards

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Code, logs, and proofs: <https://github.com/linjiw/curriculum-maxrl>

Abstract

Post-training with verifiable rewards increasingly relies on two data-level interventions: *difficulty curricula*, which reallocate rollout budget toward learnable prompts, and *failure recycling* (hindsight relabeling), which converts failed rollouts into verified successes of the tasks they actually achieved. Both are treated as objective-agnostic add-ons. We show they are not: the advantage estimator underneath shapes what each can contribute, in closed form. For success-conditioned (MaxRL-style) advantages, the expected absolute coefficient mass a prompt emits from N rollouts is exactly $2(\text{pass}@N - \text{pass}@1)$ — a compute-indexed learnability functional, vanishing exactly at pass rates 0 and 1 with a single interior peak set by the rollout budget alone — and the practical drop-all-fail implementation is unbiased for the truncation-order $T=N-1$ objective, not $T=N$. Coefficient mass is a surrogate (it is not gradient norm), but it is knowable before spending a rollout, and it predicts an estimator-conditioned divergence we then measure: across nine heterogeneous maze runs, every success-conditioned run grew $\text{pass}@k$ coverage and every GRPO run lost it — consistent, exploratory evidence that motivates the balanced factorial we outline. A two-seed GSM8K pilot points the same direction (teacher-deficit sign under GRPO in 2/2 seeds; the registered regression shape in 1 of 2). The same algebra flags a failure mode of recycling that we observe and replicate at three seeds on Countdown: relabels to already-reachable destinations improve mean accuracy while *losing* $\text{pass}@k$ coverage — **recycling-induced sharpening** — and it motivates the mitigation, gating relabel admission by the destination’s estimated pass rate. At the frontier tier the moderate gate returned coverage to the no-recycling baseline while keeping most of the mean gain; the tested settings give one useful operating point, not yet a monotonicity guarantee. The boundary of the safety claim is measured too: a pre-registered fifth suite whose learnable region collapses to one template stratum defeats *every* estimator. One diagnosis covers all of it: audit the estimator before shipping a curriculum or a recycler, and measure coverage alongside mean accuracy — the currency in which every failure mode here is visible.

1 Introduction

Post-training language models and control policies with verifiable rewards has a cost structure unlike supervised learning: the data is free, but *rollouts* are not. Every optimization step is preceded by sampling N completions per task, generation can dominate wall-clock (in our committed telemetry, eight generation-dominated sessions spanning 282 steps average 83% of step time; across all 1,188 mixed GSM8K/Countdown steps, including CPU-bottlenecked maze-scale sessions, the step-weighted mean is 31%), and on hard task pools most of that compute buys nothing — under uniform sampling we measure that **65–75% of rollout groups produce zero learning signal**: either every rollout fails (the group is dropped by success-conditioned estimators) or every rollout succeeds (no contrast, no gradient).

The field’s response is a pair of data-level interventions. The first is the *difficulty curriculum*: reallocate the rollout budget toward tasks where learning is possible right now — UCB bandits over difficulty buckets [8], target-difficulty controllers [9], category bandits [10], learnability-based rejection sampling [11], dead-prompt filtering [12, 14]. The second, newer, is *failure recycling*: convert a failed rollout into a verified success of the task it actually achieved — hindsight relabeling, recently brought into RLVR-style loops for robot policies [23]. Both interventions are, in current practice, designed and evaluated as if they were *objective-agnostic*: bolt-on modules whose benefit is assumed to transfer across the advantage estimator they sit on, and whose cost is assumed visible in the mean-accuracy metrics they are tuned against.

This paper shows that neither assumption survives contact with the estimator’s own algebra. The organizing fact is the shape of one function. **The estimator’s expected coefficient mass $u_N(p)$ vanishes exactly at pass rates 0 and 1 and peaks once in between, tracing three regimes of difficulty: a dead zone, a learnable region, and a mastered tail. Curricula reallocate compute within the learnable region; verified relabeling is a way to create a positive update for an all-fail requested task; and relabels that land in the mastered tail buy sharpening, not signal. The estimator shapes all three.** Three questions unpack this:

Q1: What can a curriculum contribute, at most? For success-conditioned advantages, the expected coefficient mass a task emits from a group of N rollouts has a closed form: $2(\text{pass}@N - \text{pass}@1)$, a compute-indexed learnability functional whose two exact zeros anchor the difficulty regimes (§3). Mass is a surrogate for learning signal (Remark 1), but it is the part a sampler can act on before spending a rollout, and it makes the allocation *derivable* rather than heuristic: the learnable region’s location, peak, and N -scaling come from the budget you already chose, and published learnability curricula are the $N=2$ slice. It also bounds the channel. Allocation can only redistribute signal that exists: a perfect-information allocator ties our cheap posterior in end performance (§6.1), and at LLM scale a prompt-level teacher starves before it pays (§6).

Q2: When is either intervention safe? The same algebra says the answer depends on the estimator. MaxRL concentrates $(N-1)\times$ more mass than RLOO on hard tasks as $p \rightarrow 0$ and releases mastered tasks; GRPO’s variance-normalized weights keep spending on both tails. We predict and measure the resulting divergence at two scales: across the audited maze runs, every MaxRL-labeled run grows pass@8 coverage and every GRPO-labeled run loses it (§6.3; exploratory — the run pool is heterogeneous); on GSM8K, the pre-registered prediction that GRPO+teacher would be the only regressing cell landed on the registered run (§6.7). Recycling has its own failure mode, and the algebra flags it: relabels that land in the mastered tail sharpen. On Countdown arithmetic this is **recycling-induced sharpening** — exact-verifier relabeling improves mean accuracy while *losing* pass@ k coverage — the data-induced sibling of GRPO’s weight-induced collapse, formally hindsight’s marginal-distribution shift [21] composed with curation-driven self-consumption [22].

Q3: Does the algebra also supply the fix? It motivates one. A relabel to a destination the model reliably reaches sits in the mastered tail, where mass vanishes, so we gate relabel admission by the destination’s estimated pass rate: the high- p zero of the same utility that schedules sampling, read as a saturation threshold on what the recycler may admit. Across three seeds the moderate gate returns frontier-tier coverage to the no-recycling baseline while keeping most of recycling’s mean gain; the available settings (moderate at three seeds, strong at one seed under corrected code) sketch a mean-versus-coverage trade but do not yet establish a monotone dial (§6.9). Where recycling is healthy the gate is transparent: hard-destination relabels pass untouched, so the validated testbeds reproduce their results with the gate enabled. One method, no per-domain switches.

Our contributions:

1. **A closed-form curriculum** (§3): the expected coefficient mass of a prompt under success-conditioned RL is exactly $2(\text{pass}@N - \text{pass}@1)$, a compute-indexed learnability functional whose peak location and N -scaling are derived rather than tuned, and whose exact zeros at $p \in \{0, 1\}$ anchor the dead zone, the learnable region, and the mastered tail; published learnability curricula are its $N=2$ slice.
2. **A derived teacher and a characterized recycler** (§4–5): FrontierMax schedules prompts by this functional with the per-group estimator untouched, and hindsight relabeling under a verified-success contract is characterized as a verified, adaptively selected auxiliary update — on-policy for the achieved task only under an explicit distribution-matching condition (Remark 3).
3. **An estimator-conditioned coverage divergence** (§6): under identical curricula, success-conditioned runs grow pass@ k coverage while GRPO runs lose it — zero exceptions across the audited maze cohort, with the same sign at LLM scale in a two-seed pilot; we grade this evidence as exploratory-but-consistent and specify the balanced factorial that would confirm it.
4. **A new failure mode and its derived mitigation** (§6.8–6.9): exact-verifier recycling improves mean accuracy while losing coverage — recycling-induced sharpening, replicated at three seeds — and gating relabels by destination pass rate yields an operating point that restores frontier-tier coverage while keeping most of the mean gain.

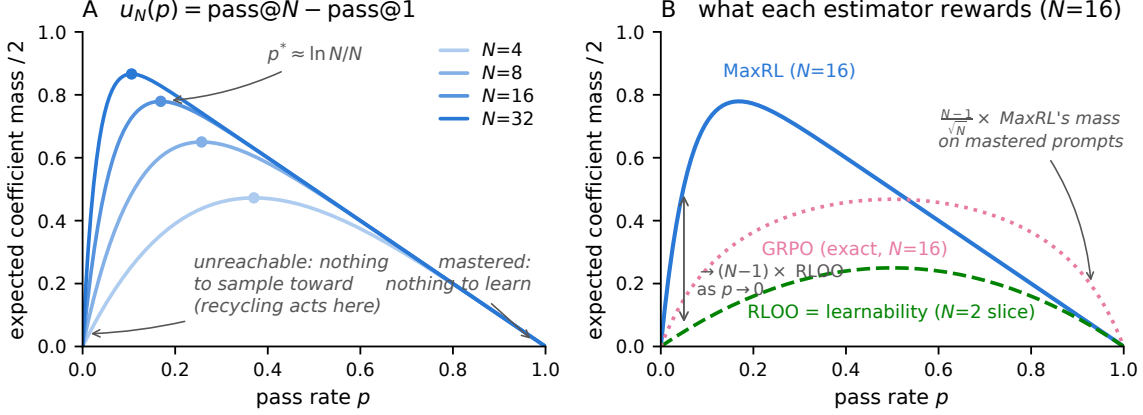


Figure 1: **The estimator is the curriculum.** *Left:* the derived utility $u_N(p) = (1 - (1 - p)^N) - p$ — half the estimator’s expected coefficient mass $A_N = 2u_N$ (Prop. 1) — for growing rollout budgets N . The peak $p^* \approx \ln N/N$ walks toward harder prompts as compute grows; the utility vanishes exactly at $p \in \{0, 1\}$. *Right:* all three curves are exact expected half-mass at $N=16$ for the estimators *as deployed* (GRPO with sample-SD normalization: $\sqrt{\frac{N-1}{N}} \frac{1}{N} \mathbb{E} \sqrt{K(N-K)}$, $K \sim \text{Bin}(N, p)$; MC-verified against the code). MaxRL concentrates $\rightarrow (N-1) \times$ more mass than RLOO on frontier prompts (Prop. 2) and releases mastered prompts fastest; GRPO’s variance normalization keeps $\sim (N-1)/\sqrt{N} \times$ MaxRL’s relative mass on mastered prompts — the maintenance-of-easy-tasks its curriculum arms lose in §6. RLOO’s profile is exactly the published “learnability” objective — the $N=2$ slice.

Along the way we state a refinement to the MaxRL paper — the practical drop- $K=0$ estimator is unbiased for the $T=N-1$ objective, not $T=N$ (Lemma 1) — and every prediction below was committed before the deciding runs, with negatives and retractions reported in place (§8) and audited in the repository.

2 Background: MaxRL in three lines

For binary rewards, maximum likelihood maximizes $\mathbb{E}[\log p_\theta(\text{success} \mid x)]$. MaxRL expands $\log p$ as a Maclaurin series in $(1 - p)$ and truncates at order T , giving a compute-indexed family $J^{(T)}$ interpolating REINFORCE ($T=1$) to exact ML ($T \rightarrow \infty$); the practical estimator sets per-rollout advantages $w_i = r_i/K - 1/N$ with all-fail groups dropped (Lemma 1 gives its exact truncation order). Its weight function grows as $p \rightarrow 0$: the objective is an *implicit, gradient-level* curriculum. Our work is the data-level complement: the same algebra, read as a sampling rule plus a recycling rule.

3 The estimator is the curriculum

Lemma 1 (Truncation order of the practical estimator). *Let $J_T(p) = -\sum_{t=1}^T (1-p)^t/t$ denote the T -truncated ML objective, so $\nabla J_T = (\sum_{j=0}^{T-1} (1-p)^j) \nabla p$; $T \rightarrow \infty$ recovers exact ML and $T=1$ REINFORCE. Under N i.i.d. binary rewards with pass rate p , the drop- $K=0$ variant of the practical MaxRL estimator has expectation $(\frac{1-(1-p)^N}{p} - (1-p)^{N-1}) \nabla p = (\sum_{j=0}^{N-2} (1-p)^j) \nabla p$: it is exactly unbiased for J_{N-1} , not J_N , for $N \geq 2$. Zeroing the control variate only on all-fail groups makes it outcome-dependent, contributing the $-(1-p)^{N-1} \nabla p$ correction, and $\sum_{k=1}^N (1-p)^{k-1} - (1-p)^{N-1} = \sum_{k=1}^{N-1} (1-p)^{k-1}$.*

Interpretation. A refinement to Tajwar et al. [1], immaterial for their conclusions but load-bearing for our mass accounting at small N — all identities below are for the estimator as actually shipped. Magnitude at the deployed budgets: the relative gradient shortfall of $T=N-1$ vs. $T=N$ is $(1-p)^{N-1} / \sum_{k=1}^N (1-p)^{k-1}$ — at $N=16$ it is 5.8% at $p=.01$, 1.1% at the band peak $p^* \approx .17$, and $< 10^{-4}$ at $p=.5$; at $N=8$ it reaches 12% on the deep frontier. Bookkeeping at moderate p , material exactly where the frontier teacher concentrates sampling.

Proposition 1 (Expected coefficient mass). *For a prompt with pass rate p and N i.i.d. rollouts under the practical MaxRL weights, define the utility $u_N(p) := \text{pass@}N(p) - \text{pass@}1(p) = (1 - (1 - p)^N) - p$. Then the expected absolute coefficient mass is*

$$A_N(p) := \mathbb{E}\left[\sum_i |w_i|\right] = 2(\text{pass@}N(p) - \text{pass@}1(p)) = 2u_N(p). \quad (1)$$

Proof (three lines): on $\{K \geq 1\}$, $\sum_i |w_i| = K(\frac{1}{K} - \frac{1}{N}) + \frac{N-K}{N} = 2(1 - \frac{K}{N})$, and $\mathbb{E}[2(1 - \frac{K}{N})\mathbf{1}\{K \geq 1\}] = 2(\Pr[K \geq 1] - p) = 2(\text{pass@}N - \text{pass@}1)$. The sampling utility used everywhere below (figures, Algorithm 1, code) is u_N ; the factor two matters only when quoting exact mass, and we carry it explicitly as $A_N = 2u_N$. Two exact companions tie mass to the gradient. Factorization: writing $\mu_{\pm}(x) = \mathbb{E}[S \mid r=1/0, x]$ for the success/failure-conditioned score means, exchangeability gives, for every group size,

$$\mathbb{E}[\hat{g} \mid x] = u_N(p) (\mu_+(x) - \mu_-(x)), \quad (2)$$

so u_N is exactly the estimator-controlled scalar multiplying the prompt’s success–failure score contrast — the part of the expected update a sampler can know before rolling out; the contrast (and where it transfers) is the part it cannot. Weight view: since $\mu_+ - \mu_- = \nabla p/(p(1-p))$, Eq. (2) is equivalent to $u_N(p) = p(1-p) w_{N-1}(p)$ with $w_T(p) = (1 - (1-p)^T)/p$ the truncated weight function of Lemma 1 — mass and the practical estimator’s weight view are linked, not identical objects. (Both identities MC-verified; artifact `verify_guidance_math.json`.)

Interpretation. *The estimator’s expected coefficient mass on a prompt is twice the probability that the prompt is solvable within N attempts but not within one. Coefficient mass is a surrogate for learning signal (Remark 1), but its two zeros are exact, and they anchor three regimes of difficulty: a dead zone at $p = 0$ (nothing to condition on; under the current policy and support, sampling alone cannot produce a success there — and at small $p > 0$ a task is operationally dead at budget (B, N) when its live-group probability $1 - (1 - \ell_N(p))^B$, with $\ell_N(p) = 1 - p^N - (1 - p)^N$ and B its share of groups, is too small to sustain training: on the frontier-heavy pool of §6.2 ($p \leq 10^{-5}$, ~ 133 groups per task) that probability is $\leq 2\%$ per task, and the one live group observed across all uniform seeds did not ignite — dead in the budgeted sense, not by fiat), a learnable region with a unique interior peak at $p^* = 1 - N^{-1/(N-1)} \approx \ln N/N$ (unique because $u_N'' = -N(N-1)(1-p)^{N-2} < 0$), and a mastered tail at $p = 1$ (no contrast left). The regimes are continuous — the only exact zeros are $p \in \{0, 1\}$ — so where we quote a “band” we mean an explicit threshold set $\{p : u_N(p) \geq \eta u_N(p^*)\}$ ($\eta=0.5$ unless stated), whose location and N -scaling follow from the one parameter you already chose. Everything in this paper is one of three moves against this structure: allocate within the learnable region (§4), create signal at its bottom (§5), and refuse updates that land in the mastered tail (the gate).*

Proposition 2 (Frontier amplification). *As $p \rightarrow 0$, the ratio of MaxRL’s expected mass to RLOO’s ($A_N^{\text{RLOO}} = 2p(1-p)$ exactly, under the $1/N$ -normalized leave-one-out weights) tends to $N - 1$.*

Interpretation. *The comparison to GRPO is two-sided and exact for the deployed implementation, which normalizes by the **sample** standard deviation ($s_K = \sqrt{K(N-K)/(N(N-1))}$, PyTorch’s default), not the population SD: $A_N^{\text{GRPO}} = 2\sqrt{\frac{N-1}{N}} \frac{1}{N} \mathbb{E}\sqrt{K(N-K)}$ with $K \sim \text{Bin}(N, p)$ (MC-verified against `estimators.py`). Taking tails, MaxRL/GRPO $\rightarrow \sqrt{N}$ as $p \rightarrow 0$ while GRPO/MaxRL $\rightarrow (N-1)/\sqrt{N}$ as $p \rightarrow 1$ (at $N=16$: $4.0\times$ and $3.75\times$; the symmetric $\sqrt{N-1}$ holds only under population-SD normalization). MaxRL over-serves the frontier and releases mastered prompts; GRPO silently keeps $\sim (N-1)/\sqrt{N} \times$ more of its mass on mastered prompts — maintenance a curriculum then removes. Concentrating sampling on the frontier only helps if the estimator extracts signal there, and §6 shows the mastered-tail difference becoming an active failure under a curriculum.*

Proposition 3 (Allocation, static). *Consider the static problem $\max_{\{N_i\}} \sum_i u_{N_i}(p_i)$ subject to $\sum_i N_i = B$ and integer $N_i \geq 0$, with pass rates p_i fixed and known and one group per prompt. Because the marginal gain $u_{N+1}(p) - u_N(p) = p(1-p)^N$ is decreasing in N , greedy water-filling on this marginal — the probability that the next rollout is a group’s first success — is optimal.*

Interpretation. *This is a mass-optimal reference allocator for the static, known- p problem only; it does not prove that sampling prompts $\propto u^\gamma$ is optimal (at fixed group size, expected immediate mass is linear in the sampling distribution, so the unconstrained myopic optimum is a brittle arg-max), nor does it describe the §6.1 oracle, which samples prompts by true- p utility at fixed group size rather than varying N_i . The practical teacher (§4) keeps fixed group sizes and samples prompts $\propto u^\gamma$ with a floor, trading myopic optimality for group-parallel batching and exploration. We measure what the gap costs: a **true-pass-rate utility oracle** (no floor, γ -matched) ties the Thompson teacher in end performance (§6.1).*

Remark 1 (Scope of the surrogate). Coefficient mass counts weighted rollouts, not score-vector geometry. We use it as the sampling utility because (i) it is knowable *before* spending rollouts, (ii) its zeros are exact — at $p \in \{0, 1\}$ the estimator emits no gradient of any norm — and (iii) on the exact-gradient testbed it *ties* the true first-order expected improvement as a task ranking (Spearman within .005 at every snapshot, and exactly 1.0 against the full exact transfer matrix — one expected group per task, exact Δp on every other task — at three training snapshots; artifacts `results_bridge.json`, `results_transfer_matrix.json`). That tie is a property of *shared-prefix* task structure, where activity and transfer align; on pools without it, activity ranks what a task can emit, not where it helps (§6.5–6.6 measure that gap). What it does not claim to be is *the* optimal sequential utility, and no state-local functional is: on independent tasks the true final-performance optimum is a planning solution (trajectory-knapsack with a budget-dependent funding cutoff), and which surrogate best approximates it depends on the training objective — a variance-tilted $(1-p)^{\alpha} u_N$ wins anytime performance while model-predictive replanning wins the final checkpoint, each losing the other’s metric (8–10/10 paired seeds each way; App. A). “Best curriculum utility” is ill-posed until that objective is fixed; we report both currencies throughout. The load-bearing claim is the pair of exact zeros at $p \in \{0, 1\}$, which every candidate utility shares.

Remark 2 (The curriculum changes the training distribution). Sampling prompts from an adaptive q_t instead of the data distribution ρ optimizes a reweighted objective $\mathbb{E}_{x \sim q_t}[J(x)]$: each group’s gradient is an unbiased MaxRL gradient *for its own prompt*, but the mixture is tilted toward the frontier, without importance correction back to ρ . This is the standard bargain every curriculum strikes (uniform sampling makes the same choice for $q_t = \rho$), and we treat it as an empirical question, not a free lunch: all evaluations in §6 are on fixed held-out pools under ρ , so any mismatch cost is charged to the method. The safety results are precisely about when this tilt helps or hurts the ρ -measured outcome.

One identity, three literatures: learnability curricula [11, 29] are the $N=2$ slice of u_N ; DAPO-style dynamic sampling and GRESO are its avoidance shadow (cull where $u \approx 0$); HER-style relabeling [15] is its creation complement (manufacture $K > 0$ where $u = 0$ identically; §5). DisCO [2] and SC-SDPO [3] derive analogous per-question weights for the GRPO family to diagnose difficulty bias; our identity generalizes that line to success-conditioned estimators and — the step neither takes — uses it to predict how *external* curricula and recyclers interact with each estimator, in coverage terms.

The three regimes, as a map of the paper (Fig. 3). The **teacher** (§4) allocates compute within the learnable region — and saturates: a true-pass-rate oracle allocator ties the cheap posterior in end performance (§6.1), so better pass-rate information is not where remaining gains live. **Recycling** (§5) creates signal where sampling the requested task yields none — in our control battery, the one intervention that still adds on top of the oracle. The **estimator** underneath shapes whether either is safe, and its mastered tail is where both failure modes live (§6). One line: *the teacher allocates, hindsight creates, the estimator decides whether either is safe*.

4 FrontierMax: the schedule

FrontierMax is a standard group-rollout RLVR loop with three modified lines (Alg. 1). A decayed Beta posterior tracks each prompt’s pass rate from observed group outcomes only — never from relabeled successes, which inflate it ($\hat{p}$ 0.81 vs. eval 0.47). Thompson sampling draws $\tilde{p}$ and samples prompts $\propto u(\tilde{p})^{\gamma}$ with a uniform floor. The knobs (decay 0.7, floor 0.1, $\gamma \in \{1, 4\}$) carry validated defaults (Table 3); what is *derived* rather than tuned is the difficulty band itself — location, width, and N -scaling — which is exactly where competing curricula spend their hyperparameters.

5 Failure recycling

Remark 3 (Hindsight update, characterized). Relabeling a dead group to the sub-goal g' its trajectories actually achieved, rewriting the goal conditioning, and applying the same success-conditioned weights yields a *verified weighted-likelihood update for the achieved task*: every relabeled example is a true success of g' under the environment’s verifier. It is generally an *adaptively selected, off-policy auxiliary update*, not an

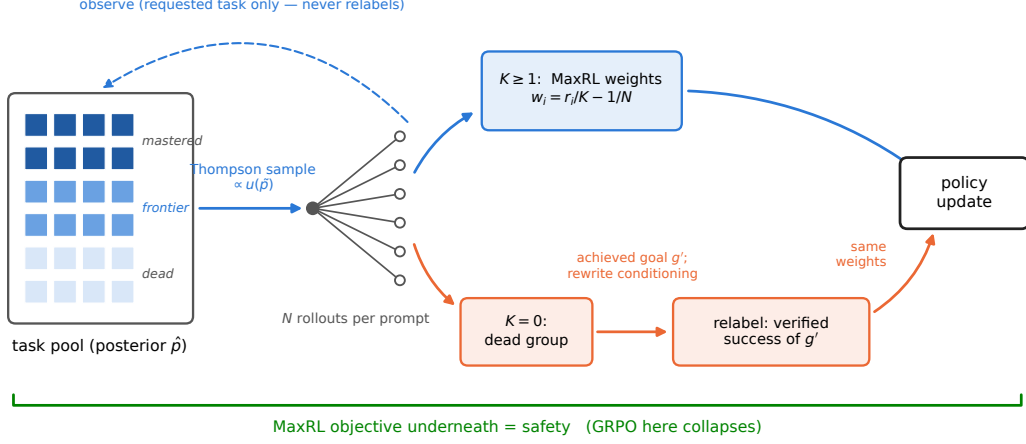


Figure 2: **One FrontierMax step.** A Thompson-sampled posterior allocates rollouts by the derived utility; live groups train with unmodified MaxRL weights; dead groups are relabeled to the sub-goal they actually achieved (conditioning rewritten) and train as verified successes. The posterior observes requested-task outcomes only.

Algorithm 1 FrontierMax: one step. Three lines differ from a standard group-rollout loop (marked ▷).

- 1: ▷ **sample** prompts $x_1..x_B \propto u_N(\tilde{p}_x)^\gamma$ mixed with a uniform floor, $\tilde{p}_x \sim \text{Beta}(\alpha_x, \beta_x)$ (else: uniform)
- 2: roll out N completions per prompt; verify; $K_x \leftarrow$ successes
- 3: update posterior (α_x, β_x) from (K_x, N) *for the requested task only*
- 4: live groups ($1 \leq K_x$): MaxRL advantages $w_i = r_i/K - 1/N$, unchanged
- 5: ▷ **dead groups** ($K_x=0$): relabel each parseable failure to *its own* achieved sub-goal g' (verified; conditioning rewritten); unparseable rollouts stay failures, so the group retains contrast ($K < N$)
- 6: ▷ admit the relabel only if $\hat{p}_{g'} \leq \text{gate_max_p}$ (the gate: refuse the mastered tail)
- 7: one policy-gradient step on live + admitted relabeled groups

unbiased gradient of g' 's on-policy objective: g' is selected from the same group being scored, conditioning on the requested task's failure correlates the rollouts, the conditioning rewrite changes the law under which the log-probability is evaluated, no importance ratio corrects for it, and the finite- N weights target J_{N-1} (Lemma 1) rather than exact ML. It equals the fresh-rollout J_{N-1} gradient for g' only under a distribution-matching condition — the selected, rewritten group must have the same joint law as an i.i.d. fresh rollout group for g' . This specializes known relabeling-bias structure — the induced-distribution view of Eysenbach et al. [20] and the conditional-law gap in Ghosh et al. [21] — to success-conditioned RLVR weights; we state it as a remark, not a result, and note that the sharpening phenomenon of §6.8 runs through the *marginal* shift those works also identify, not this conditional matching.

Interpretation. “Exactness” is a property to measure per environment, not a blanket guarantee. On our validation testbed the laws match by construction, so cosine probes there are consistency checks, not evidence (several null relabeling schemes produce identical cosines); the evidence that the relabel direction carries information is behavioral — the placebo decomposition below. We measured the two gaps the remark names where they are measurable (exact chains, 5 seeds; artifact `results-proposal-shift.json`): the applied relabel update’s cosine against the exact fresh-destination gradient direction is 0.93, the relabeled group’s success rate exceeds the true fresh destination pass rate by only +.003 (selection guarantees $K' \geq 1$, so the bias exists but is small here), and cross-fitting the destination choice (select on half the group, update with the other half) removes the adaptive reuse at a real cost: alignment drops to 0.82 and AUC falls .883 $\rightarrow$.829, because the selected successes leave the update half. The off-policy caveat is real; its sign is environment-specific.

Two contracts keep practice on the proof’s side: (1) **exactness** — a relabeled success must be a true success of the relabeled task under the environment’s own verifier (never an LLM judge); (2) **conditioning** — goal-embedding trajectories must have their conditioning rewritten to the achieved goal. Skipping (2) turns

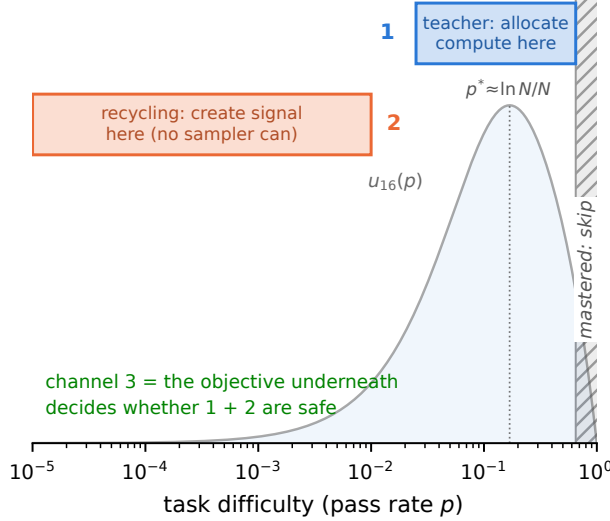


Figure 3: **The three regimes as a map of the method.** The teacher allocates compute within the learnable region; recycling creates signal where sampling the requested task yields none; the gate refuses relabels that land in the mastered tail; the estimator underneath decides whether any of it is safe.

the exact gradient into noise: on our gridworld, hindsight without the rewrite scores below teacher-only ($0.600 < 0.658$); with it, it leads (0.703). Among the interventions studied here, recycling is the one that *creates* signal for an all-fail requested task (adding the achieved sub-goals to the pool by hand would too — what recycling buys is discovering them, §6.2), and we decompose its headline gain rather than let it inflate: of the $+0.22$ AUC on fixed task pools, a placebo replaying *live* gradients on dead-group slots captures 83% — most of the benefit is extra gradient dose on otherwise-wasted slots — while the relabel *direction* carries $+0.037$ beyond the strongest placebo, is the tightest arm in the battery, and is exactness-dependent (random-target relabels lose to dose-matched replay 5/5). Both numbers ship together. The value is regime-dependent in a way we can state precisely: $+0.22$ AUC on fixed task pools, $+0.01$ on one-shot task streams — the gain is proportional to how much a relabeled skill can compound, and task-set fixedness is the single most important regime variable we found.

6 Experiments: an escalating ladder

Each rung asks one question of the three-regime structure, at increasing scale and decreasing experimental control; Table 1 is the map. Statistical contract, stated once and used throughout: the independent unit for any training-method claim is an independently trained seed block; runs sharing a seed, warmstart, or checkpoint are paired or correlated observations, never additional replicates; levels, k values, metrics, and checkpoints from one trained model are repeated measurements of it; repeated-evaluation spread measures evaluation noise, not training-seed uncertainty, and every $\pm$ names its source at point of use ($\pm$ with no qualifier is standard deviation across training seeds). “Coverage” is pass@ k with exactly k samples drawn per prompt and averaged over the eval pool, at the k named per suite (8 on the maze, 16 on Countdown, 4 on GSM8K); mean@ k is the mean success fraction of the same k samples; AUC is the mean of the eval curve over training at matched wall-clock or episode budget, and final is the last checkpoint. Where run pools are heterogeneous we say so and report signs descriptively rather than attaching p -values (§6.3–6.4). Pre-registered predictions were committed before the deciding runs; negative results appear in place, one sentence each, with full postmortems in the repository’s audit trail.

6.1 Does allocation saturate? (exact-gradient skill chains; CPU, 5 seeds). Uniform $0.650 \rightarrow$ teacher $0.728 \rightarrow$ full stack 0.890 ± 0.002 AUC. An eight-arm control battery (App. A) orders into a *gradient-information ladder*, each rung a 5/5 paired-seed win: zero-information dose (0.798 – 0.800) $<$ replayed

Table 1: The ladder at a glance: every rung asks one question of the three-regime structure; scale rises, control falls.

rung	question	setting	seeds	outcome
6.1	does allocation saturate?	skill chain (exact grads)	5	yes: oracle ties posterior
6.2	anything to sample toward?	frontier-heavy pool	5	no: only creation scores
6.3	main effect, real grads?	maze transformer, 1.26M	9 runs	9/9 signs (exploratory)
6.4	where/when does coverage go?	maze, level-resolved	22 runs	easy band, immediately
6.5	when does allocation not pay?	IsaacLab Anymal-C	1	learnable-everywhere: null
6.6	beyond task spread?	MountainCar/CartPole	3	spread=existence; tea.=speed
6.7	LLM scale?	GSM8K 2×2, 360M	1–2	sign landed; replication open
6.8	what does recycling cost?	Countdown v2	3	mean up, coverage down
6.9	does the gate recover it?	Countdown v2	3	one useful trade point

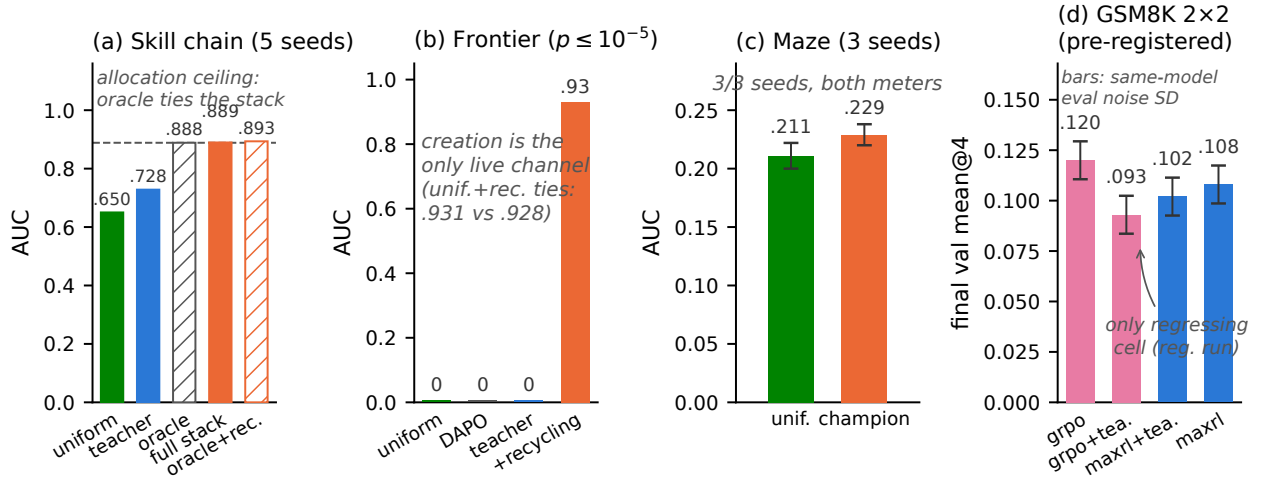


Figure 4: **The ladder.** (a) Exact-gradient chains: a no-floor, γ -matched true-pass-rate oracle matches the full stack within the observed five-seed variation — allocation saturates — and recycling still adds on top of the oracle. (b) Frontier-heavy regime: allocation scores exactly zero; recycling solves it, and does so equally under uniform sampling (.931 vs. .928) — creation, not allocation, is the live channel. (c) Maze at matched wall-clock: 3/3 seeds on both meters. (d) GSM8K 2×2, registered run, error bars = measured same-model eval noise (SD .0094): GRPO+teacher is the only cell that regresses (pre-registered P-G2; a replication seed did not reproduce it — §6.7); the teacher’s MaxRL effect is a null at this budget (P-G1).

correct direction (0.832) < true relabels (0.869) < perfect allocation (oracle 0.8885 $\approx$ full stack 0.8895) < allocation+creation (0.8935). Two decompositions matter downstream. *Allocation saturates*: a no-floor, γ -matched true-pass-rate oracle (0.8885 $\pm$.0014) sits within the five-seed variation of the full stack (0.8895 $\pm$.0023) — similar within observed noise, not tested for equivalence — so perfect difficulty information is worth little over the cheap posterior. *Creation decomposes*: a replay placebo captures 83% of recycling’s +0.22; the relabel *direction* carries +0.037 beyond it and is the most reliable arm in the battery, while random-target relabels lose to dose-matched replay 5/5. The utility form is load-bearing through its N : the N -blind learnability slice $p(1-p)$ forfeits the entire allocation gain (−.007 vs. uniform) where u_N carries +.078 (5/5). All arms converge within .027 of one another — on fixed pools these interventions buy *time-to-ceiling*, not the ceiling.

Takeaway. A monotone information ladder: dose < direction < allocation-ceiling < +creation. Allocation saturates; creation adds on top; every intervention buys speed, not asymptote.

6.2 Can any sampler act when nothing is learnable? (frontier-heavy regime, max pool pass rate 10^{-5} ; 5 seeds, artifact committed). Uniform, DAPO, and the plain teacher score **exactly 0.00**; adding

recycling reaches **0.98 final (0.93 AUC)** — relabeling invents the curriculum below the pool, ignites within ~ 400 groups, then goes silent. The attribution control pins down which channel did it: *uniform*+recycling scores the same ($0.931 \pm .002$ vs. teacher+recycling $0.928 \pm .003$ AUC) — in this regime the entire effect is signal creation; allocation contributes nothing because there is nothing learnable to allocate toward. Equivalently: a designer who hand-added the achieved sub-goals to the pool would also escape the zero — what recycling buys is *discovering* that sub-pool from the policy’s own failures, with no task engineering. One loss-level alternative remains: the MaxRL paper’s own Eq. (10) keeps the control-variate term $-\frac{1}{N} \sum_i S_i$ on all-fail groups, so that estimator is *not* silent at $K=0$ ($\mathbb{E}[\hat{g} \mid K=0] = \nabla p / (1-p)$, MC-verified). We ran it (5 seeds, matched budget; artifact `results_fullcv_baseline.json`): on this pool the full-CV variant spends 100% of its update norm on all-fail groups and still scores 0.000 in every seed, with or without the teacher — the all-fail term pushes down every sampled action symmetrically and cannot ignite a pool this deep, while recycling under either variant reaches 0.98. On the balanced pool the two variants tie (paired Δ AUC straddles zero, 5 seeds).

Takeaway. Where nothing is learnable at budget, sampling policy is irrelevant — uniform and teacher tie at zero without recycling and tie at 0.93 with it. Even the estimator family’s nonzero all-fail update (full control variate) stays at zero here: creation is the only channel we tested that ignites a dead pool, and its value is discovery, not allocation.

6.3 Does the estimator main effect survive real gradients? (maze transformer at matched wall-clock; 1.26M params, 13 levels, ~ 30 runs, 3 seeds). Champion (teacher + dense recycling) 0.252 ± 0.005 final / 0.229 ± 0.009 AUC vs. uniform 0.230 ± 0.015 / 0.211 ± 0.011 ; paired deltas positive in 3/3 seeds on both meters (final and AUC — six correlated contrasts, not six seeds); 22–35% more optimization steps per GPU-hour. Attribution within that gain, stated plainly: matched *wall-clock* lets curriculum arms take $1.2\text{--}1.4\times$ more optimization steps (dead groups skip the backward pass), and step-matched truncation shows the extra steps are most of the pure teacher’s effect (step-matched teacher-only Δ AUC $+0.006$, within seed noise) while teacher+recycling survives step-matching (positive in all three paired seed blocks; a paired t at $n=3$ gives $p \approx .04$, which we report as descriptive given its normality sensitivity — an exact sign-flip test cannot go below .25 at three blocks) — on the maze the teacher’s measured value is *throughput* (buying more steps per GPU-hour), recycling’s is signal. The maze teacher uses the legacy frontier form; a pre-registered 3-seed rerun with the exact u_N measures the two within noise *on the maze itself* (AUC $.2138 \pm .0059$ vs. $.2135 \pm .0104$; artifact `un_form_verdicts.json`), closing the gap between the deployed form and the derived one. The same sweep refutes a CPU-derived within-band refinement (a $(1-p)$ -tilted utility that won 10/10 on the testbed loses 0/3 on the maze on every meter) — within-band utility choices are testbed-sensitive, which is why the theory claims only the band’s boundaries. Safety, stated as what the design supports: an estimator-associated coverage difference — all five MaxRL-labeled runs grow pass@8 (e.g. $0.316 \rightarrow 0.348$ under the teacher) and all four GRPO-labeled runs lose it ($0.308 \rightarrow 0.271$ uniform; $0.332 \rightarrow 0.269$ with the teacher). We state this as what it is: perfect sign separation across nine runs whose pool mixes sampler configurations (the MaxRL five span frontier and uniform teachers; the GRPO four span uniform and one frontier) and reuses seed blocks and warmstarts, so the runs are not exchangeable and a label-permutation p -value would answer a randomization this design did not perform — we therefore report zero exceptions descriptively and do not attach one. Two sensitivity reads support the same direction without curing the design (artifact `permutation_reanalysis.json`): restricting permutations within matched sampler arms preserves the perfect separation, and all three shared-warmstart pairs are individually positive; the run composition (5v4 because one uniform-MaxRL seed predates pass@ k logging) is tabulated in the same artifact. The confirmatory design this motivates is a balanced $\{\text{MaxRL}, \text{GRPO}\} \times \{\text{uniform}, \text{teacher}\}$ factorial over at least six independent seed blocks with identical warmstarts and one prespecified endpoint — fewer than six blocks cannot reach two-sided $p < .05$ even with unanimous signs. At the exact-gradient rung, where the design is cheap, we already ran its schedule-matched core: freeze each seed’s *realized* prompt sequence (one recorded teacher trajectory, one uniform), then replay the identical sequence under each estimator from the same init — the estimator ordering on exact summed pass@8 survives with the schedule held fixed (MaxRL over GRPO in 10/10 seed $\times$ schedule pairs; artifact `results_schedule_matched.json`), with the caveat that on that learnable-everywhere pool GRPO still gains coverage — the neural-scale *loss* is what the balanced maze factorial must test. Whether the teacher *amplifies* GRPO’s decay is a factorial (interaction)

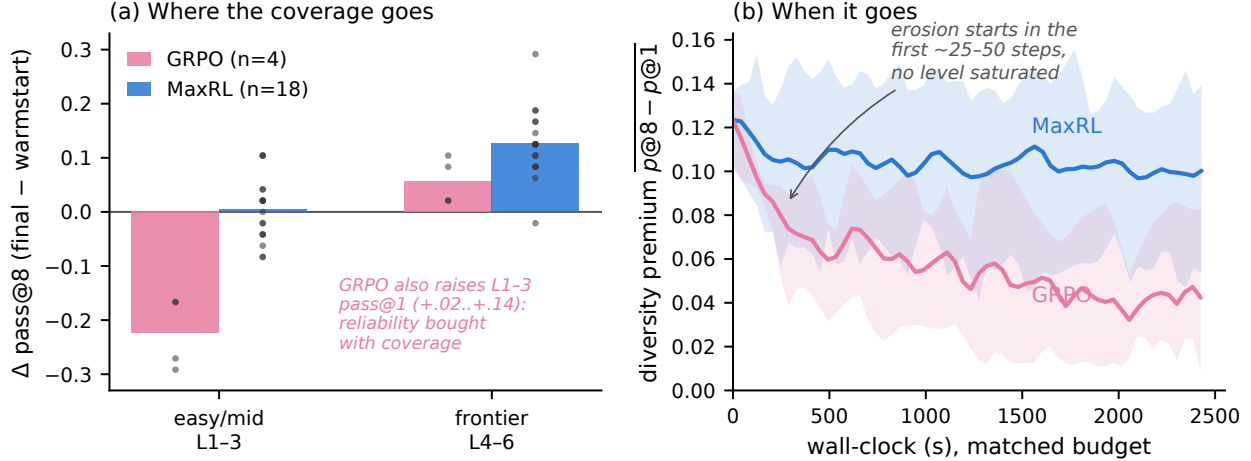


Figure 5: **Where and when GRPO’s coverage goes (maze, level-resolved, matched wall-clock).** (a) $\Delta \text{pass@8}$ (final – warmstart) by difficulty band: GRPO’s loss lives on the easy/mid band (all 4 runs; dots are runs) while its pass@1 there rises — sharpening; MaxRL holds the easy band and gains on the frontier band (18 runs; a heterogeneous run inventory — descriptive, no p -value). (b) Diversity premium $\text{pass@8} - \text{pass@1}$ vs. wall-clock (mean and min–max across runs): GRPO’s erosion begins within the first ~25–50 steps, before any level saturates; MaxRL stabilizes after warmup.

question this suite is not powered for; the GSM8K 2×2 below tests exactly that cell, pre-registered. Inference currency (one convention, every level: smallest k reaching absolute coverage 0.25; single checkpoint pair): our checkpoint needs $11 \times$ fewer samples at level 5, ties GRPO at levels 2–3, and needs $3 \times$ more at level 4 — the frontier is where the efficiency gain lives. The summed-coverage view agrees without a target choice: GRPO ahead at pass@1, curves crossing at $k \approx 4$, MaxRL ahead by half a level at $k=64$ (App. A) — choose the estimator by how many samples you will draw at inference.

6.4 Where and when does GRPO’s coverage go? (level-resolved). Decomposing the main effect by difficulty band turns it into a mechanism (Fig. 5). *Where:* GRPO’s pass@8 loss is concentrated almost entirely on the easy/mid band L1–3 ($-.224$ mean over its four runs, every run negative) while its L1–3 pass@1 *rises* ($+.02$ to $+.14$, every run) — it funds reliability by liquidating easy-band coverage; MaxRL holds L1–3 flat ($+.005 \pm .058$ over 18 runs) and concentrates its gain on the frontier band L4–6 ($+.126 \pm .063$ vs. GRPO’s $+.057$). The 18-run MaxRL pool is a heterogeneous run inventory (teacher variants, hindsight doses, a wide model, repeated seeds), so we report the band asymmetry descriptively — every GRPO run negative on L1–3, the MaxRL spread centered at zero there — and attach no permutation p -value; the 22 runs are not exchangeable replicates of one estimator contrast. This is consistent with the mass curves in Fig. 1 — the $\sqrt{K(N-K)}$ weighting keeps spending updates on mastered prompts where u_N has already released them — but the step from mass to coverage is a stated *hypothesis*, not a derivation: mass is sign-blind, and “mastered-tail updates sharpen” requires an argument about the sign structure of near-saturated groups (a lone failure draws a large negative weight under variance normalization, pushing mass off the failure mode and onto the dominant one) that we support with the location and timing evidence here rather than prove. *When:* the divergence is immediate, not saturation-triggered — GRPO incurs 59–73% of its total diversity-premium loss ($\text{pass@8} - \text{pass@1}$: $.123 \rightarrow .042$) within the first quarter of the budget, with the first easy-band pass@8 drop visible at steps 0–50 when L1 pass@1 is still only $.62-.75$; MaxRL’s premium stabilizes after warmup ($.122 \rightarrow .101$). A dose-response note from the same logs (descriptive, same caveat as above): sparse relabeling (1/dead group) uniquely preserves the diversity premium relative to none and dense, while dense banks the same coverage into the best final pass@1 — dose sets *where* the coverage dividend is spent, not whether coverage is lost.

Takeaway. The gains survive real gradients, and the coverage main effect now has a located mechanism: GRPO liquidates easy-band coverage for reliability from the first updates; MaxRL reallocates the same budget to the frontier. Coverage is the currency that sees it.

6.5 When does allocation *not* pay? (IsaacLab legged locomotion, pre-registered null). A 5-arm AnyMal-C rough-terrain pilot (one seed, run by an independent team on a co-designed protocol; App. A) lands its pre-registered null exactly: on a terrain grid where every level is learnable at budget, the stock greedy curriculum beats the frontier teacher outright (mean pass .372 vs. .278). Read with the maze step-matched analysis and Countdown, the consistent regime pattern — at one seed here, so a pilot-level observation rather than proof — is: **allocation pays where unlearnable-at-budget regions exist to avoid**; on learnable-everywhere pools, breadth beat focus in every case we measured.

6.6 What does a curriculum add beyond task spread? (Gymnasium control — binary-success RLVR on classic control; official-task currency). All arms train on *binary task success* (MountainCar: reach the flag; CartPole: survive 400) — the classic-control analogue of a verifier reward. The **target-only** arm is then a provable zero-update freeze, not a slow learner: every group is all-fail, every group-relative estimator emits exactly zero weights on constant rewards, and telemetry confirms zero parameter updates over the full budget (random-policy success $< 3 \times 10^{-6}$; no budget ignites it). The gated full stack under the identical reward reaches **1.000±0.000 on both official tasks** (3 seeds, Fig. 8). A dense-reward control confirms the zero is a property of sparse verifier-style reward (REINFORCE on CartPole’s native reward solves it; App. A). Attribution at a common horizon: uniform-over-bins task spread supplies most of the effect (existence of a gradient), recycling-without-allocation plateaus below ceiling, and the teacher’s remaining contribution is speed and stability ($3 \times$ faster to hard-task 0.9, $2.5\text{--}6 \times$ smaller cross-seed variance). A methods note we keep because it generalizes: our own first version used per-bin task-partitioned policy parameters and reproduced the known failure (0.000 in every arm) — curricula operate through shared parameters, or not at all.

Takeaway. Curricula operate through shared parameters, or not at all.

6.7 Does it transfer to LLM scale? (GSM8K 2×2 , pre-registered; SmolLM2-360M, $N=16$, one A10G). Predictions committed before any cell finished. **P-G2 landed on the registered run:** GRPO+teacher was the only cell that regressed in its second half, and the deficit against plain GRPO at the endpoint is $2\text{--}4 \times$ the measured same-model eval noise (mean@4 $-.027$; pass@4 $-.048$; the second-half dip itself, $.096 \rightarrow .093$, is within the $\pm .009$ noise floor and carries no weight on its own) while plain GRPO climbed monotonically ($.078 \rightarrow .105 \rightarrow .120$). One uncertainty caveat governs every GSM8K number: the $\pm .009/\pm .017$ bars are repeated-eval noise of a fixed checkpoint — they bound evaluation variability, not training-seed variability — so deficit-to-noise ratios are descriptive signal-to-noise reads, not hypothesis tests; training-seed uncertainty at $n=2$ is what the queued multi-seed cell measures. Steering telemetry, read against its own controls: the teacher’s sampled-dead fraction *minimum* reaches .48 vs. uniform’s .52, but the run means are indistinguishable (.66–.67 in all cells) — the delivered treatment was weak, which is the posterior-starvation mechanism below, and it bounds how much of the GRPO gap the curriculum can own. **A replication seed does not reproduce the regression shape:** seed 2 of the same cell climbs monotonically ($.073 \rightarrow .095 \rightarrow .118$), and its steering telemetry shows the treatment was weaker (min dead-sampled fraction .53, mean .66 $\approx$ population — a near-uniform teacher on average). The completed *within-seed* pair (grpo-uniform seed 2: $.081 \rightarrow .085 \rightarrow .125$, final pass@4 .229; artifact `grpo_uniform_seed2.json`) sharpens the read: the endpoint teacher deficit under GRPO has the same *sign* in both seeds on both meters (mean@4 and pass@4 are computed from the same generations and strongly correlated, so this is two seed-level confirmations read on two coupled meters, not four; seed 2: $-.007$ mean@4, within noise, $-.026$ pass@4, $\sim 1.5 \times$ its floor) with magnitude tracking steering intensity — a dose effect, not a sign flip. The pre-registered *second-half-regression* shape remains 1 of 2 seeds: suggestive, not established. MaxRL+teacher trains stably to its best value ($.066 \rightarrow .102$); the uniform-MaxRL cell finishes at .108, making P-G1 (a teacher gain for MaxRL) a *null at this budget* — predicted in advance by the posterior-starvation analysis below. A pass@ k sweep on the registered checkpoints: the teacher’s deficit under GRPO *widens monotonically with k* ($-.008$ at $k=1$ to $-.024$ at $k=16$) — curriculum damage lives in coverage, as the maze predicted, though each individual

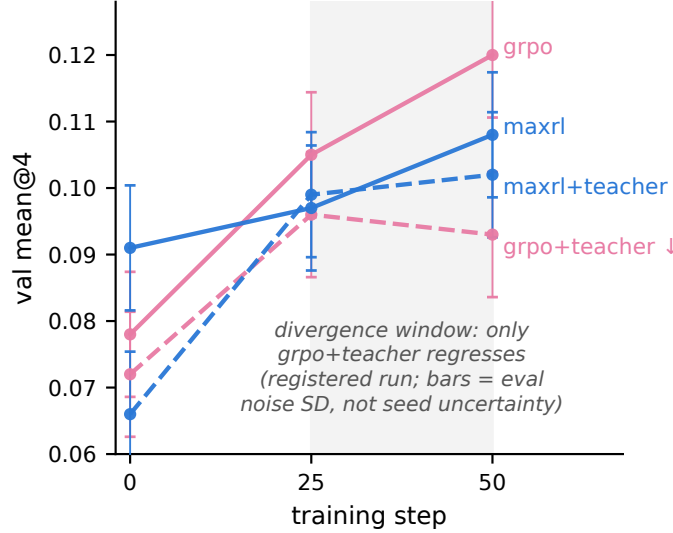


Figure 6: **GSM8K 2×2 (pre-registered; registered run shown; error bars = measured same-model eval noise, SD .0094)**. GRPO+teacher is the only cell that regresses in its second half (P-G2 on this run; the load-bearing statistic is the endpoint gap vs. plain GRPO, 2–4× the repeated-eval noise — a descriptive signal-to-noise read, not a test; the second-half dip alone is within noise; a replication seed with weaker measured steering climbed instead — §6.7); under MaxRL the identical teacher trains stably to its best value but does not beat uniform at this budget (P-G1: null).

delta is within single-seed noise (harness caveats in App. A). One design confound is disclosed rather than dismissed: the teacher samples with replacement, so curriculum cells revisit prompts more — a repetition effect is an alternative reading that a uniform-with-replacement control cell separates; it is running with the steering-controlled replication. The mechanism behind the weak treatment: the teacher’s posterior *learns* real difficulty from ~ 1 visit per prompt ($\rho = -0.17$ against independent 7B-model solve-rate annotations, $p \approx 10^{-17}$; cross-run posterior agreement 3× its agreement with the external key — it learns *this model’s* landscape) but its allocation is *posterior-starved*: 3,200 draws over 7,473 prompts leave the estimated utility landscape nearly flat, so Thompson sampling stays near-uniform (full posterior diagnostics in App. A). Small model; the pre-registered outcomes were orderings and signs on the registered run.

Takeaway. The maze sign pattern stands on nine heterogeneous runs; the LLM-scale interaction is 1-of-2-seeds and tracks steering intensity — the decisive experiment is a steering-controlled multi-seed cell, queued.

6.8 What does recycling cost? (Countdown 2×2, pre-registered): two nulls and a new phenomenon. The first exact-verifier hindsight experiment in RLVR ran in production (12 relabeled groups/step, 72–89% yield). Both positive predictions returned null, with an exact diagnosis: the uniform baseline reached 94/76/94% of the random-guesser bound implied by the pool’s simplified construction — no headroom for allocation or creation to matter. What the run found instead is new: **recycling-induced sharpening** — hindsight arms improved mean@16 while *losing* pass@16 coverage (tier-1: .571 vs. .645 for the no-relabel baseline). Recycled off-target successes concentrate the policy on reachable outputs at the expense of exploration breadth: the creation channel has its own safety trade-off, mirroring §6.3’s weight-induced collapse — data-induced rather than weight-induced. The underlying concentration pathology is known in goal-conditioned RL (Skew-Fit’s motivating failure mode; GCSL’s marginal shift); what is unreported, to our knowledge, is its measurement in RLVR — no prior work quantifies the pass@ k cost of hindsight relabeling in a verifier loop. Our algebra flags the mechanism: the achieved-goal marginal overweights destinations the policy already reaches, so relabel updates concentrate toward the mastered tail, where $u(p)$ says fresh training signal vanishes — the softmax pays for those updates out of the exploration tail. (One selected success does not by itself imply the destination’s fresh-rollout pass rate is high; that is what the

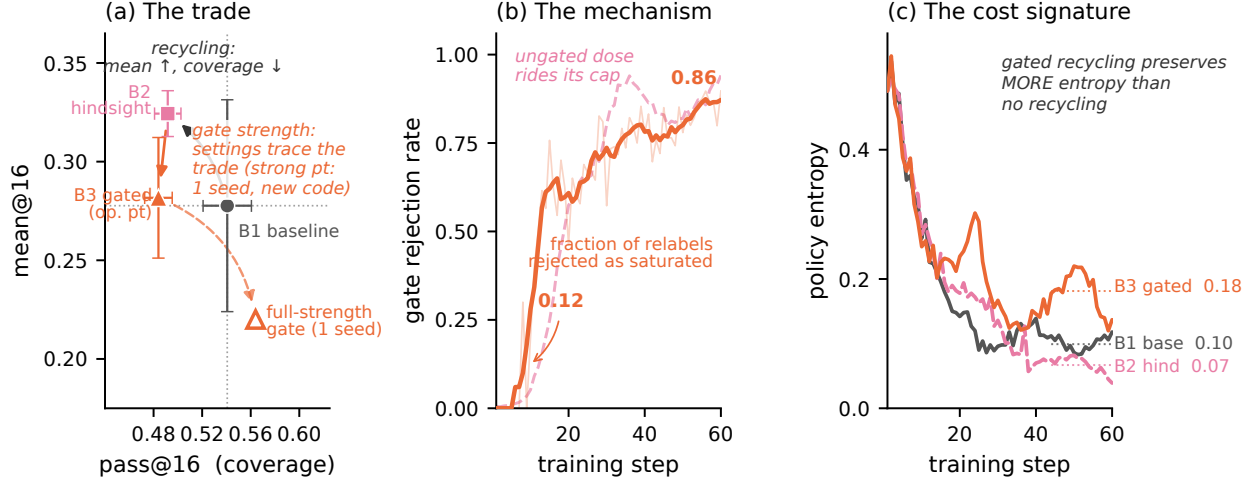


Figure 7: **Recycling-induced sharpening and its derived mitigation (Countdown, tier 1).** (a) Three seeds, both axes: recycling (B2) buys mean@16 with pass@16 coverage. On this tier the moderate gate (B3) gives back most of the mean without moving coverage — tier-1 destinations saturate early, so admitted relabels are few — while full-strength gating (corrected decay, 1 seed) returns coverage *above* baseline at the mean’s full price. The settings are consistent with a mean-vs-coverage trade (not yet a monotone dial: the strong point is one seed on different code), and the useful middle lives on the frontier tier, where the same moderate setting restores coverage to baseline (.279±.019 vs. .274) keeping ~60% of the mean gain (§6.8). (b) The mechanism in-run: the gate’s rejection rate rises monotonically as the model’s reachable set saturates its relabel destinations, while the ungated dose rides its cap. (c) Gated recycling preserves *more* policy entropy than no recycling — admitted relabels are a spread-preserving gradient.

gate’s independent destination estimate $\hat{p}_g$ checks.) *Takeaway: recycling needs an admission rule, and the utility’s high- p zero motivates one — a destination-saturation threshold on the estimated pass rate $\hat{p}_g$ (the implementation thresholds $\hat{p}$ directly; it does not evaluate u_N).*

6.9 Does the gate recover it? (Countdown v2 pool, 3 seeds, pre-registered). We rebuilt the pool with real structure (random operand permutation, parenthesization, exact division; a headroom gate excluded guesser-saturated tiers) and ran three arms $\times$ three seeds: no recycling (B1), ungated recycling (B2), gated recycling (B3). Setup note with a theory reading: recycling has a *syntactic prerequisite* — the pre-SFT base model produces zero verifier-valid failures (relabel yield 0.000, $n=384$), so there is nothing to recycle until SFT installs the output format; post-SFT the model is a near-independent guesser whose pass@16 sits at a tier-invariant ~75% of the independence bound $1 - (1 - \text{pass@1})^{16}$ (ratios .744/.783/.755 across tiers) — the baseline our headroom gate measures against. **Sharpening replicates:** at tier 1, recycling buys mean@16 +.046 (.278±.054 $\rightarrow$.324±.012) and pays -.049 of pass@16 (.541±.020 $\rightarrow$.492±.011). The coverage loss clears both arms’ seed spreads ($> 2\times$); the mean gain clears the treated arm’s tight spread but sits within the noisier baseline’s — we report it as directionally consistent across seeds rather than individually significant. The frontier tier moves the same direction. **The timing is dose-then-damage:** recycling’s frontier-tier mean gain is installed by step 30 — +.014±.008 paired, all seeds positive, at only ~28% of the eventual relabel dose — while the coverage cost only materializes by step 45 (−.048±.019 paired, *all seeds negative*, after ~65% of dose): the benefit precedes the damage, so early-stopping on mean-accuracy validation would bank the gain and never see the bill. The damage is also visible on the training side: the ungated arm ends with more unsolvable-at-16 frontier training prompts than baseline in every seed; the gate’s entropy retention replicates across seeds while ungated recycling makes entropy seed-unstable (details in App. A). **The gate works where the algebra says it should:** at the frontier tier (destinations not yet saturated) the gated arm returns coverage to baseline (.279±.019 vs. .274±.015) while keeping ~60% of the mean gain; at tier

1 (destinations saturate early, so the gate blocks nearly all relabels) it neutralizes recycling almost entirely. Implementation disclosure: a decay-math bug made all three-seed gated runs gate *weaker* than designed (effective destination threshold $\hat{p} \approx 0.64$ where the design implies 0.92), so the reported operating point is an under-gated one; the corrected-decay rerun (1 seed) gates $\sim 3\times$ harder and pushes tier-1 coverage *above* baseline at the mean’s full price (Fig. 7a). The three settings are *consistent with* a mean-vs-coverage trade — no gate: full mean gain, coverage lost; under-gated: $\sim 60\%$ of the gain, frontier coverage restored; full-strength: gain erased, coverage above baseline — but they do not establish a monotone dose-response: the strong point is one seed, gate setting is confounded with the decay-code version, and at tier 1 the moderate gate moved *both* point estimates below the ungated arm. We claim the moderate frontier-tier setting as one useful operating point and defer the dial claim to the pre-registered fixed-decay sweep. In-run telemetry shows the mechanism (Fig. 7b–c): rejection rises $.12 \rightarrow .85$ as the reachable set saturates, and the gated arm ends with *more* policy entropy than the no-recycling baseline. A pre-registered `gate_max_p` dose sweep at fixed decay is the standing follow-up.

Takeaway. Sharpening is a reproducible property of exact-verifier recycling, and the derived gate buys back frontier-tier coverage at a measured mean price — one validated operating point, with the monotone-dial question deferred to a fixed-code sweep.

7 Related work

The coverage tension. RLVR itself trades coverage for precision: `pass@1` rises while `pass@k` at large k falls below the base model [4], with entropy collapse as the mechanistic signature [6] and overtraining on saturated problems as a locus [5]. Our contribution to this line is the *interaction*: whether external curricula and recyclers compound or cancel the estimator’s implicit weighting, and what that costs in coverage. Notably, recent work frames curricula as the *cure* for coverage loss [7]; our GRPO arms show the cure is estimator-conditional — those studies never varied the estimator. The estimator-difficulty interaction itself is a heating 2026 theme — SC-SDPO [3] shows GRPO’s normalization implicitly encodes a difficulty weighting, and A-GRAE [28] redesigns the advantage around difficulty focus — but no prior work runs *identical* curricula under two estimators or measures the resulting `pass@k` sign flip.

Curricula for RLVR. Difficulty-adaptive sampling is bucket- or scalar-level and heuristic-scored: DUMP (UCB over buckets), AdaRFT (scalar controller over precomputed labels, TMLR 2026), SEC (category bandit), threshold curricula. LILO (NeurIPS 2025) is the closest per-prompt method — rejection sampling by $\hat{p}(1 - \hat{p})$, which Prop. 1’s corollary identifies as the $N=2$ slice of our derived utility. Reinforce-Ada [13] — the adaptive sampler MaxRL itself points to — adapts the rollout count *within* a prompt until a signal criterion is met, holding the prompt distribution fixed; our teacher is the complementary move, reallocating *across* prompts at fixed group size, and Prop. 3 is the idealized form of both (water-filling spends its next rollout wherever $p(1 - p)^N$ is largest, across or within). DAPO’s dynamic sampling and GRESO cull dead prompts — avoidance without allocation or creation. PKPO and Pass@k-training modify the objective toward `pass@k` but keep uniform sampling. **Goal selection in HER.** Selecting *which* goals to train on by difficulty or reachability is a decade-old idea in the relabeling literature: intermediate-difficulty goal generation [16], value-based hindsight goal selection (HGG; 17), curriculum-guided HER’s proximity-diversity admission of relabel goals [18], and Skew-Fit’s α as a concentration-vs-coverage dial [19]. Our gate belongs to this mechanism class; what is new is not admission-by-difficulty but its *derivation* — the threshold is the mastered-tail zero of the same estimator functional u_N that schedules sampling, so teacher and gate are one object with no separately tuned goal-selection heuristic — and the coverage accounting it is validated against.

Hindsight and self-training for LLMs. Exact-verifier relabeling has precedent outside RLVR loops — CodeIt [24] (ARC program synthesis, relabel + prioritized replay) and proof-tree relabeling [25] — and LLM-judged variants relabel agent trajectories (AgentHER, HSL). Concurrent 2026 work brings relabeling-adjacent mechanisms into RLVR-style training: LfH [23] (a VLM proposes one shared hindsight instruction per failed group for VLA post-training; the relabeler is a judge, their stated limitation) and ZipRL’s hindsight response replay [26] (densifying GRPO signal for multi-turn agents). None of these measures `pass@k` under relabeling — precisely the currency where we find recycling’s hidden cost. The nearest phenomenon report is in same-problem rejection finetuning: Strozzi [27] document a STaR-trained model winning `pass@8` while the base model overtakes at `pass@256` — concentration over reach — for SFT on a model’s own verified outputs.

Recycling-induced sharpening is the cross-task, on-policy-RL sibling, and our gate is its estimator-derived admission rule; gating *vanilla* RLVR updates by saturation signals has independent precedent (5’s Bayesian boundary gating), not applied to relabel destinations. So the precise claims we retain: no prior work (i) derives the sampling utility from the estimator’s own expected signal, (ii) couples it with success-conditioned weights it provably matches, or (iii) *measures the coverage cost of hindsight relabeling in RLVR* and gates relabel admission by destination saturation.

8 Limitations and negative results

Scale. The LLM-scale teacher×estimator interaction is 1-of-2 seeds and tracks steering intensity: the replication seed of the GRPO+teacher cell did not reproduce the registered regression, and its telemetry shows the treatment was barely delivered (§6.7). The GSM8K teacher is also coverage-neutral-to-negative under *both* estimators at this budget, so the contrast there concerns the sign of the mean-accuracy effect, not a coverage rescue. The maze coverage divergence (nine runs, zero sign exceptions, but a heterogeneous pool — §6.3) is the strongest evidence this paper has for the safety claim, and it is exploratory; the balanced maze factorial and a steering-controlled multi-seed LLM cell are the decisive next experiments.

Scope of the surrogate. Coefficient mass is not gradient norm, SNR, or expected improvement (Remark 1) — it omits score-vector geometry, sequence length, token masking, optimizer state, and clipping — and §6.5 exhibits the regime where reallocating it buys nothing (learnable-everywhere pools). Recycled-gradient exactness is a property of the environment’s relabel map; noisier verifiers will land between our fixed-pool and one-shot endpoints (+0.22 vs. +0.01 AUC).

The coverage main effect is pool-conditional. A pre-registered fifth suite (Jugs water-measuring puzzles; preregistration, per-cell raw endpoints, verdicts, and postmortem vendored under `jugs_llm/results/` with execution-commit provenance) returned all-null: on a pool whose learnable band is a single stratum of 1–2-move template solutions, *plain MaxRL* also collapses — entropy $1.36 \rightarrow 0.07\text{--}0.21$ in 60 steps, frontier-tier $\text{pass@16 } 0.15 \rightarrow 0$, every arm converging to the same deterministic policy — and recycling accelerates the collapse rather than rescuing it. When N rollouts of a near-trivial plan are near-identical, the i.i.d.-rollout diversity the group contrast requires is absent, and success-conditioned weights liquidate frontier coverage through shared parameters just as GRPO’s do elsewhere. The estimator main effect of §6.3–6.4 is therefore a claim about pools with a graded band at the deployed N , not an unconditional safety guarantee; per-task rollout-set diversity at initialization is the design gate we now check first. The same suite also exposed a granularity bug in the gate (keying destinations by value collides tasks when achieved values are few and shared — ≤ 20 integers in Jugs vs. hundreds in Countdown); the corrected gate keys on the relabeled task.

What did not transfer. A $4\times$ duration run refuted our own duration hypothesis (the deep-frontier stall is a per-step-competence ceiling; depth needs capacity, not schedule); γ -concentration did not transfer from chains to the maze, as the compounding model predicted; adaptive truncation order underperformed fixed T ; and feeding relabeled successes to the teacher’s posterior inflates it, so the posterior sees requested-task evidence only.

Corrections along the way. Two early claims were retracted on internal review — a teacher-steering read traced to an epoch-frozen sampler (the fixed run confirmed the effect), and a “beats the oracle” comparison traced to a floor handicap in the oracle arm (§6.1 reports the corrected match-within-noise). Full postmortems live in the repository’s audit trail.

9 Conclusion

For the practical success-conditioned estimator under i.i.d. binary rewards, the expected absolute coefficient mass of a prompt is exactly $2(\text{pass}@N - \text{pass}@1)$, and the drop-all-fail implementation targets truncation order $N-1$. That identity supplies a compute-aware sampling heuristic and an admission threshold for verified

relabels; it does not by itself prove a learning band, optimal prompt allocation, or on-policy hindsight, and we have been explicit about where each stronger reading fails. Empirically, estimator choice and relabel admission changed the trade between mean accuracy and pass@ k coverage everywhere we looked: an exploratory but zero-exception coverage divergence between MaxRL- and GRPO-labeled maze runs, a three-seed replication of recycling-induced sharpening on Countdown with one gated operating point that buys the coverage back, and a pre-registered negative (Jugs) marking where the whole family of interventions fails together. The confirmatory experiments these results specify — a balanced estimator-by-sampler factorial at six or more seed blocks, a fixed-code gate-strength sweep, and a steering-controlled multi-seed LLM cell — are pre-registered and queued. The portable practice we advocate is independent of our method: measure coverage next to mean accuracy whenever a curriculum or recycler ships, conditioned on the estimator it ships on.

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A Full details for compressed experiment rungs

Material moved from §6 during the page-budget pass; every number here is artifact-backed in the repository.

6.1 control battery (full). Eight arms, 5 seeds each: uniform 0.650, lr $\times$ 2 0.798, random-target relabels 0.800, live-gradient replay 0.832, true relabels 0.869, no-floor γ -matched oracle $0.8885\pm.0014$, full stack $0.8895\pm.0023$, oracle+recycling $0.8935\pm.0012$. The relabel direction term (+.037 over replay) has seed spread $\pm.0025$, 4–6 $\times$ tighter than every placebo; its worst seed beats every placebo’s best. Random-target relabels are harmful relative to dose-matched replay ($-.032$, 5/5 seeds) but neutral vs. plain lr-doubling: fake labels waste the slot rather than poison training. Utility-form controls: exact u_N $0.728\pm.002$ vs. legacy frontier form $0.733\pm.013$ (the derived form’s payoff is 7 $\times$ smaller seed variance); the $N=2$ slice $p(1-p)$ is a wash against uniform ($-.007$, 2/5 wins) where u_N carries +.078 (5/5). AUC spread across arms is 9 $\times$ the final-performance spread.

6.5 IsaacLab protocol. Five arms: frontier teacher, stock greedy walker, uniform, scripted ramp, no-motion control; fixed-grid per-level evaluation; Anymal-C rough terrain; one seed, executed by an independent team against a co-designed, pre-registered protocol. The teacher’s only edge over the greedy walker is a within-noise sliver at the hardest evaluated level.

6.6 gym control (full). MountainCar flag: measured random-policy success $< 3\times 10^{-6}$ over 10^6 episodes; survive-400 $\approx 10^{-13}$ by tail fit. Dense-reward controls (prediction registered before the run): CartPole’s native +1/step reward lets vanilla REINFORCE on the same tile policy solve survive-400 (0.96 ± 0.03 , 3 seeds); MountainCar’s native -1 /step reward (constant until first success) stays at 0.000 in all seeds — the categorical zero tracks reward sparsity, not the environment. Attribution at a common 280-step horizon (hard-task AUC): MC .750 $\rightarrow$ +recycling .895 $\rightarrow$ +teacher .956; CartPole .708 $\rightarrow$.792 $\rightarrow$.893; paired teacher increment positive 3/3 seeds on both; creation-without-allocation plateaus at 0.958 on CartPole in every seed while adding the teacher restores 1.000; teacher speed effect 3.0 $\times$ to hard-task 0.9 (2.1 $\times$ to 0.99 on CartPole). The per-bin ablation replicates the 10-seed transition-matched study (flag 0.848 ± 0.058) before strengthening it to full convergence.

6.3 inference-efficiency detail. Convention: smallest k reaching absolute coverage 0.25, log-interpolated; artifact `maze_gpu/efficiency.json`; Fig. 9 plots the summed-coverage crossing. Summed coverage over L0–6: GRPO 3.24 vs. MaxRL 3.07 at pass@1; crossing at $k\approx 4$; 5.12 vs. 4.62 at $k=64$, with GRPO’s level-2/4 curves saturating at 0.88/0.56 where MaxRL’s reach 1.00/0.62.

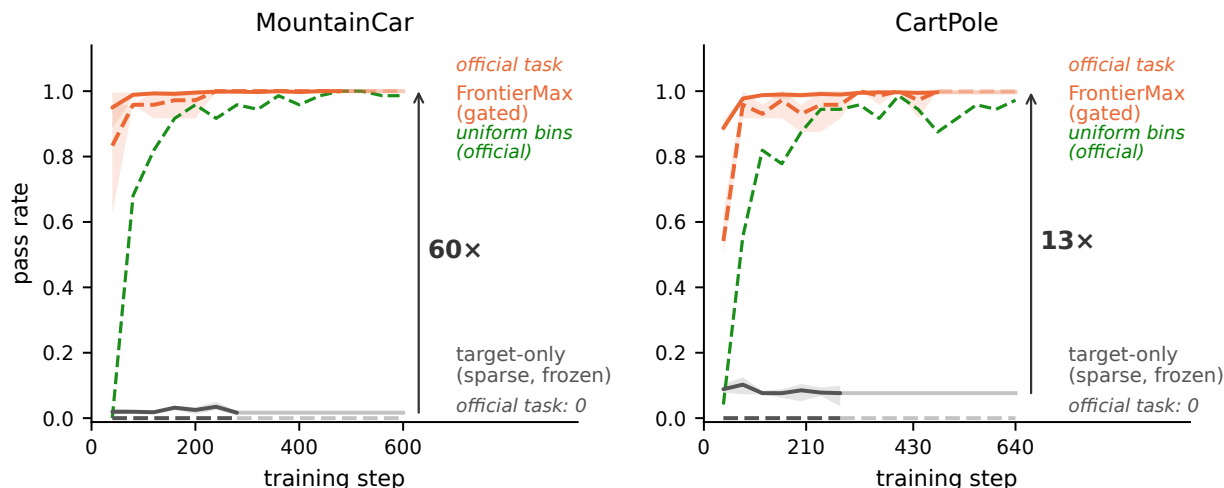


Figure 8: **Gym control under binary task-success reward, trained to plateau (3 seeds).** Solid: mean pass rate across curriculum bins; dashed: the *official* task (MountainCar flag / CartPole survive-400). The target-only arm (gray) is a provable zero-update freeze under sparse verifier-style reward; a uniform-over-bins control (green) also cracks the task, 3 $\times$ slower and off-ceiling; the gated FrontierMax stack converges to 1.000 ± 0.000 .

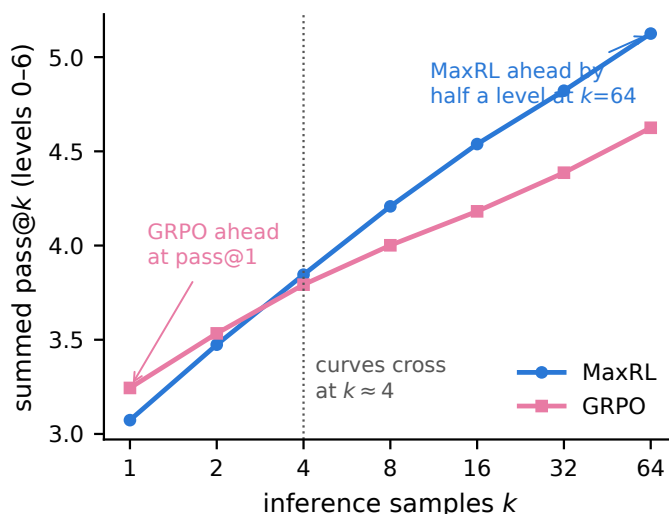


Figure 9: **Summed maze coverage vs. inference samples k (final uniform-arm checkpoints).** GRPO leads at pass@1; the curves cross at $k \approx 4$; MaxRL leads by half a level at $k=64$ — choose the estimator by how many samples you will draw at inference.

6.9 secondary reads (full). Training-side damage: ungated recycling ends with more unsolvable-at-16 frontier training prompts than baseline in every seed (paired $+0.083 / +0.063 / +0.034$). Entropy at 3 seeds: the gate’s retention replicates ($+0.072$ paired over baseline, all seeds positive, $\sim 1.6\times$); ungated recycling’s endpoint entropy is seed-unstable (spread ± 0.062 , 4–9 $\times$ the other arms’), consistent with its noisy frontier-coverage endpoints.

6.7 posterior diagnostics and harness caveats (full). The pass@ k sweep’s absolute level sits $\sim 3\times$ below the trainer’s val metric (unreconciled harness discrepancy; only the within-sweep contrast is quoted).

Posterior detail: two independent teacher runs’ posteriors agree with each other (cross-run Spearman $\approx .45$ on co-visited prompts, $n \approx 900$; $\approx .67$ self-consistency with a shared latent) vs. $.10$ – $.22$ against external 7B-model annotations, from ~ 1.25 visits per prompt. Across external-difficulty quintiles spanning solve rates $.33 \rightarrow .98$, mean $\hat{p}$ moves only $.076 \rightarrow .103$ — the compressed landscape that starves allocation. Response length separates runs by estimator, perfectly: all three GRPO runs compress to 209.6 ± 1.8 tokens while both MaxRL runs sit at 232.4 ± 5.7 (common ~ 365 -token warmstart; the teacher moves length ≤ 3 tokens) — sharpening visible in a currency where token entropy is ambiguous. Prompt-uniqueness figures for the replacement confound: $\sim 34\%$ unique prompts seen by teacher arms vs. $\sim 43\%$ uniform.

Jugs suite (pool-conditionality; §8). Tier grid t0–t4 (2–5 jugs, disjoint min-moves bands, 200 unique tasks/tier); feasibility landscape (SmolLM2-360M two-shot, $N=16$): t0 $.058/.475$ (pass@1/pass@16), t1 $.006/.100$, t2–t4 0/0 with relabel-yield-on-failure $.62$ – $.73$ on every tier. All nine pre-registered B-arm cells (no-recycling / ungated / gated $\times 3$ seeds) converged to the same deterministic fixed point: t0 mean@16 = pass@16 = 0.450 exactly (the 2-move template stratum), t1 pass@16 $0.15 \rightarrow 0$, actor entropy $1.36 \rightarrow 0.02$ – 0.21 . Ungated recycling ran 169 relabels/step at yield $.88$ and ended with the lowest entropy (0.02 – 0.03); the value-keyed gate rejected 99.8% of relabels (the granularity bug described in §8, since fixed and regression-tested). Postmortem and per-cell artifacts in the repository.

B Hyperparameters and compute

Total compute for every experiment in this paper: ~ 230 A10G-hours on a single GPU plus CPU-days for the exact-gradient suites. Step-time telemetry (artifact `generation_timing.json`, 1,188 mixed GSM8K/Countdown steps): step-weighted mean generation fraction 31% overall; the eight generation-dominated sessions ($>80\%$ session mean, 282 steps) average 83% — a selected-subset description, not an estimate of typical cost. Table 2 gives the per-suite configuration and Table 3 the knob inventory with provenance.

Table 2: Per-suite configuration. Teacher knobs (decay, floor, γ) are shared across suites with the defaults below; what varies per suite is scale, not schedule.

	chains (§6.1–6.2)	maze (§6.3–6.4)	GSM8K (§6.7)	Countdown (§6.8–6.9)
policy	exact tabular	1.26M transformer	SmolLM2-360M	SmolLM2-360M (SFT)
rollouts N	16	32	16	16
groups/step	1	8	64	64
optimizer, lr	SGD, 0.5	AdamW, 10^{-4}	AdamW, 10^{-5}	AdamW, 10^{-5}
budget	3,200 groups	2,400 s clock	50 steps	60 steps
seeds (headline)	5–10	3	1–2/cell	3/arm
compute/run	CPU-min	0.7 A10G-h	~ 20 A10G-h	~ 6 A10G-h

Table 3: Teacher and recycler knobs: value, provenance, and measured sensitivity.

knob	default	derived or tuned	sensitivity
peak location/ N -scaling	—	derived (Prop. 1)	the theory claim
posterior decay	0.7	tuned (V2b)	closes $\sim 19\%$ of oracle gap
uniform floor	0.1	tuned	0.15 on maze; robust 0.05–0.2
γ (concentration)	1	tuned (V6)	4 on chains; no transfer to maze
<code>gate_max_p</code>	0.5	$p \rightarrow 1$ zero, midpoint tuned	one trade point validated (§6.9)
relabel dose cap	12 groups/step	tuned	sets <i>where</i> coverage is spent (§6.4)

Reproducibility

Every proposition has an enumeration/Monte-Carlo verification script (`test_mass_formulas.py` in `curriculum_maxrl/` checks the MaxRL, RLOO, and both GRPO-SD mass formulas against exact enumeration and the deployed code). Every figure regenerates from committed inputs listed, with checksums and commands, in the frozen manifest `paper/results/manifest.json`: result values live in versioned JSON tables under `paper/figures/data/` (none are hard-coded in figure scripts), and the maze cohort of Fig. 5 is frozen in an explicit run manifest — adding a run file cannot silently change the figure, and missing inputs are a hard error. The Jugs preregistration, raw cells, verdicts, and postmortem are vendored in `jugs_llm/results/` with preregistration- and execution-commit provenance. Known gaps, stated rather than papered over: the IsaacLab pilot is reported from an independent team’s summary (raw logs live with that team); per-step timing fractions come from verl’s in-run telemetry (`generation_timing.json`); the fig2/fig3 endpoint tables are transcribed from run analyses rather than derived from a seed-level results table, and building that table is queued. The framework is a dependency-light package (`frontier_rl`, numpy-only core) with adapters for LLM pools (verl), gym control, IsaacLab parallel sim, and flow-policy VLAs.